

# A fast and robust two-point ray tracing method in layered vertical transversely isotropic media with strong anisotropy

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## Abstract

A fast and robust two-point ray tracing method was developed for layered vertical transversely isotropic media with strong anisotropy. Utilizing the Christoffel slowness equation, a novel generalized dimensionless ray parameter,  $q_x$ , modified from the ray parameter (horizontal slowness), was proposed to efficiently and simultaneously determine the ray paths and travel times for direct and reflected quasi-P, quasi-SV and quasi-SH waves. The Newton optimization algorithm was employed to solve the non-linear offset equation accurately, resulting in rapid convergence to the true value. The inferred analytical equations show that the generalized ray parameter stabilizes the inversion process at large offsets. Additionally, a piecewise function was introduced to enhance the initial value estimation and calculation efficiency. The numerical results demonstrate that this novel approach can reduce the iteration error to  $10^{-10}$  m in less than three iterations. Monte Carlo simulations further validated the effectiveness of the  $q_x$  method for inferring the true ray paths at various offsets within complex velocity models. Furthermore, the  $q_x$  method can address the triplication issue in quasi-SV waves and exhibit robustness in strong-layered vertical transversely isotropic media.

## KEYWORDS

Anisotropy, Numerical study, Rays, Seismics, Theory

## INTRODUCTION

Ray tracing, including shooting methods and bending methods, have been widely acknowledged as effective approaches for determining ray paths and travel times between seismic sources and receivers in various geophysical applications (Bóna et al., 2009; Browning et al., 2013; Bulant, 1996; Fang & Chen, 2019; Prothero et al., 1988; Um & Thurber, 1987; Wang, 2014; Xu et al., 2006). In seismology, the

ray tracing problem in three-dimensional (3D) formations is typically simplified to approximate solutions using layered models (Thurber & Ellsworth, 1980). The two-point ray tracing method, which calculates the ray path from source to receiver, is essential for tasks such as earthquake location, travel time tomography, migration, velocity model optimization and numerical simulations. Therefore, there is an ongoing demand for a fast and reliable two-point ray tracing method that efficiently addresses iterative problems

(Asakawa & Kawanaka, 1993; Julian & Gubbins, 1977; Nakanishi & Yamaguchi, 1986; Vidale, 1988; Vinje et al., 1993).

Layered vertical transversely isotropic (VTI) media have been extensively utilized in seismic pre-stack migration (Bakulin et al., 2010; Shi et al., 2023) and microseismic monitoring of shale hydraulic fracturing in exploration geophysics (Grechka & Yaskovich, 2014; Jarillo Michel & Tsvankin, 2017; Li et al., 2013; Yaskovich et al., 2020; Zhang et al., 2022). The accuracy of the final image relies heavily on precise calculations of ray paths and travel times in seismic wave propagation. In VTI media, three types of waves, quasi-P (qP), quasi-SV (qSV) and quasi-SH (qSH), travel in specific directions with varying phase and group velocities (Shearer & Orcutt, 1985). Identifying group velocities plays a crucial role in determining accurate ray paths and poses challenges owing to their complex nature. This study specifically focused on determining the precise ray paths in highly anisotropic layered VTI media using a two-point ray tracing method.

Thomsen (1986) pioneered the characterization of phase and group velocities using five Thomsen parameters. The discrepancy in the angles between the two velocities necessitates a multistep calculation procedure for ray paths and travel times. Thomsen suggested simplifying the group velocities because of the weak anisotropy in most rocks. Several approximate analytical equations, such as those proposed by Byun et al. (1989) and Sena (1991), provide rapid inference of ray paths for group velocities. They perform inadequately in strong anisotropic formations (Faria & Stoffa, 1994; Kim et al., 2021; Zhang et al., 2022). Hence, the assumption of weak anisotropy is unsuitable for VTI media with strong anisotropy.

The qP, qSV and qSH waves in strongly anisotropic media can be expressed using the ray parameter in the Christoffel slowness equation. Han et al. (2017) utilized the Christoffel slowness surface of the qP wave and proposed a numerical quartic equation solver in two-dimensional (2D) Tilted Transversely Isotropic media. Bouteiller et al. (2018) presented a discontinuous Galerkin method for accurately and robustly solving the Eikonal equation in 2D heterogeneous anisotropic media. Lu and Zhang (2021, 2022) further calculated the travel times of 2D qSV/qSH and 3D qP waves using a quadratic Christoffel slowness equation transformation. The Christoffel slowness equation is well suited for ray-path calculations in strongly anisotropic VTI media; however, the computation-intensive problem leads to low efficiency in the petroleum industry.

The generalized ray parameter method, first proposed by Tian and Chen (2005), has the potential to rapidly calculate ray paths in layered VTI media with strong anisotropy. This method replaced the ray parameter in the two-point ray tracing method of Kim and Baag (2002) with a modified one and quickly solved the offset equation using the Newton iterative

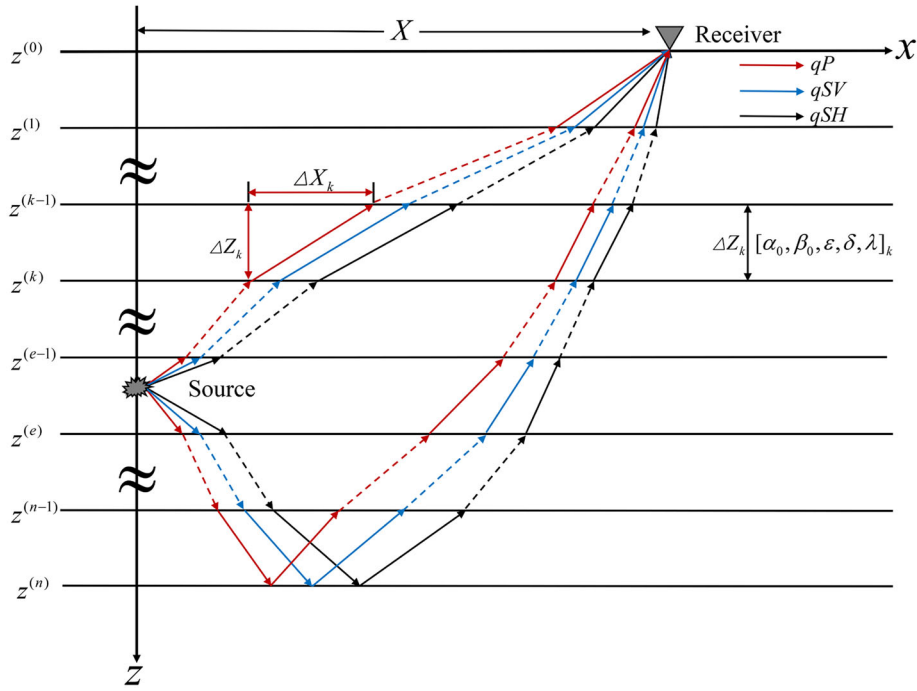
method. It can rapidly and stably converge to the true ray paths at any offset in the isotropic constant-layered model. Fang and Chen (2019) further extended the generalized two-point ray parameter ray tracing theory to horizontally layered media, with the layer velocity being constant or varying linearly with depth. They confirmed the robustness and efficiency of their method, particularly for large offsets. The generalized ray parameter method has been successfully employed to solve imaging challenges in oil exploitation, as demonstrated by Pei et al. (2009), Belayouni et al. (2015) and Zhang et al. (2019). However, these applications have been limited to isotropic media, highlighting the need for further research and development of anisotropic media exploration and analysis.

In this study, we extended the generalized ray parameter method to strongly anisotropic layered VTI media. Detailed formulae based on the Christoffel slowness equation were derived, and the Newton method was used to efficiently solve the offset equation. An initial estimate of the ray parameters was applied to enhance the effectiveness, and the travel times and ray paths of both the direct and reflected waves were determined accurately.

The rest of this paper is organized as follows. First, we provide a brief overview of the ray parameter method based on the Christoffel slowness equation in VTI media. Second, we introduce a modified generalized ray parameter to address the convergence issues encountered at distant offsets. We then propose a piecewise function to accurately estimate the initial values of the generalized ray parameter. Subsequently, we present several examples to demonstrate the efficiency and accuracy of the proposed method. Finally, the advantages and primary applications of the generalized ray parameter method are summarized.

## METHODOLOGY

In layered vertical transversely isotropic (VTI) media, both direct and reflected waves play important roles in seismic imaging. The ray paths of these waves are shown in Figure 1.  $z^{(k)}$  represents the  $k$ th layer depth and  $\Delta Z_k$  is the  $k$ th layer thickness. The  $k$ th layer anisotropy intensity is described using five Thomsen parameters:  $[\alpha_0, \beta_0, \epsilon, \delta, \gamma]^k$ . For direct waves, quasi-P (qP), quasi-SV (qSV) and quasi-SH (qSH) waves are excited at the source in the  $e$ th layer, then traveling upward and reaching the receiver at the surface. Conversely, the reflected waves first travel downward and then reach the receiver on the surface after reflecting off the  $n$ th layer interface. The directions of the ray paths change in each layer according to Snell's law (Aki & Richards, 1980). The  $k$ th layer horizontal ray path is denoted as  $\Delta X_k$ , where  $X$  represents the horizontal offset between the source and receiver. The qP, qSV and qSH waves follow distinct ray paths, resulting in varying travel times.



**FIGURE 1** A schematic diagram illustrates the ray paths of direct waves and reflected waves in layered vertical transversely isotropic (VTI) media.  $z^{(k)}$  and  $\Delta Z_k$ , respectively, represent the  $k$ th layer depth and thickness.  $[\alpha_0, \beta_0, \epsilon, \delta, \gamma]^k$  are the  $k$ th layer Thomsen anisotropic parameters. One source in layer  $e$  and one receiver on the surface are shown. The ray paths of the direct waves travel from the  $e$ th layer to the surface, whereas the ray paths of the reflected wave involve reflection from the  $n$ th interface. The different coloured paths represent different wave types: red for quasi-P (qP) wave, blue for quasi-SV (qSV) wave and black for quasi-SH (qSH) wave.  $X$  denotes the horizontal offset between the source and the receiver,  $\Delta X_k$  represents the  $k$ th layer horizontal ray path.

The horizontal offset,  $\Delta X_k$ , in the  $k$ th layer is directly proportional to its corresponding horizontal group velocity component,  $V_{xk}$ . This relationship can be mathematically expressed as

$$\Delta X_k = \Delta Z_k \cdot \frac{V_{xk}}{V_{zk}}, \quad (1)$$

where  $\Delta Z_k$  denotes the  $k$ th layer thickness,  $V_{xk}$  and  $V_{zk}$  represent the horizontal and vertical group velocity components of the  $k$ th layer, respectively. The  $k$ th layer group velocity travel time,  $\Delta T_k$ , can be determined by the layer thickness,  $\Delta Z_k$ , and the vertical group velocity component,  $V_{zk}$ :

$$\Delta T_k = \frac{\Delta Z_k}{V_{zk}}. \quad (2)$$

In accordance with Christoffel's dispersion theory, the horizontal and vertical group velocities,  $V_{xk}$  and  $V_{zk}$ , can be expressed in terms of the slowness components:

$$V_{xk} = \left( p_{xk} \frac{\partial S(p_{xk}, p_{zk})}{\partial p_{xk}} + p_{zk} \frac{\partial S(p_{xk}, p_{zk})}{\partial p_{zk}} \right)^{-1} \frac{\partial S(p_{xk}, p_{zk})}{\partial p_{xk}},$$

$$V_{zk} = \left( p_{xk} \frac{\partial S(p_{xk}, p_{zk})}{\partial p_{xk}} + p_{zk} \frac{\partial S(p_{xk}, p_{zk})}{\partial p_{zk}} \right)^{-1} \frac{\partial S(p_{xk}, p_{zk})}{\partial p_{zk}}, \quad (3)$$

where  $p_{xk}$  and  $p_{zk}$  are the  $k$ th layer horizontal and vertical slowness components, respectively.  $S(p_{xk}, p_{zk})$  represents the slowness equation. In VTI media, the qP and qSV waves couple, and their respective Eikonal equations form a quartic slowness surface (Lu & Zhang, 2022):

$$S_{qP}(p_{xk}, p_{zk}) = p_{zk}^2 - \frac{B_k - \sqrt{B_k^2 - 4C_k}}{2} = p_{zk}^2 - g_1(p_{xk}) = 0,$$

$$S_{qSV}(p_{xk}, p_{zk}) = p_{zk}^2 - \frac{B_k + \sqrt{B_k^2 - 4C_k}}{2} = p_{zk}^2 - g_2(p_{xk}) = 0,$$

$$B_k = \frac{1}{\alpha_{0k}^2} + \frac{1}{\beta_{0k}^2} - 2 \left( 1 + \delta_k + (\epsilon_k - \delta_k) \frac{\alpha_{0k}^2}{\beta_{0k}^2} \right) p_{xk}^2,$$

$$C_k = \left( (1 + 2\epsilon_k) p_{xk}^2 - \frac{1}{\alpha_{0k}^2} \right) \left( p_{xk}^2 - \frac{1}{\beta_{0k}^2} \right). \quad (4)$$

In contrast, the qSH-wave slowness surface takes the form of a simple ellipse described by the following equation:

$$\begin{aligned} S_{\text{qSH}}(p_{xk}, p_{zk}) &= p_{zk}^2 + (2\gamma_k + 1) p_{xk}^2 - \frac{1}{\beta_{0k}^2} \\ &= p_{zk}^2 - g_3(p_{xk}) = 0, \end{aligned} \quad (5)$$

where  $S_{\text{qP}}(p_{xk}, p_{zk})$ ,  $S_{\text{qSV}}(p_{xk}, p_{zk})$  and  $S_{\text{qSH}}(p_{xk}, p_{zk})$  represent the surface slowness equations for the qP, qSV and qSH waves, respectively. Terms  $\alpha_{0k}$  and  $\beta_{0k}$  signify the  $k$ th layer qP- and qSV-wave velocities along the symmetry axis.  $\epsilon_k$  and  $\delta_k$  are Thomsen parameters (Thomsen, 1986; Tsvankin, 1997).  $B_k$  and  $C_k$  are the simplified variables.  $g_1(p_{xk})$ ,  $g_2(p_{xk})$  and  $g_3(p_{xk})$  are the simplified formulae of  $p_{xk}$  used to solve  $p_{zk}$  for the qP, qSV and qSH waves, respectively.

Using Equations (3)–(5),  $\Delta X_k$  can be expressed in terms of horizontal slowness through the following:

$$\begin{aligned} \Delta X_k &= \Delta Z_k \cdot \frac{V_{xk}}{V_{zk}} = \Delta Z_k \cdot \frac{\frac{\partial S(p_{xk}, p_{zk})}{\partial p_{xk}}}{\frac{\partial S(p_{xk}, p_{zk})}{\partial p_{zk}}} \\ &= -\Delta Z_k \cdot \frac{\partial \sqrt{g(p_{xk})}}{\partial p_{xk}} = -\Delta Z_k \cdot \frac{\partial G(p_{xk})}{\partial p_{xk}}, \end{aligned} \quad (6)$$

where  $S(p_{xk}, p_{zk})$  is shorthand for  $S_{\text{qP}}(p_{xk}, p_{zk})$ ,  $S_{\text{qSV}}(p_{xk}, p_{zk})$  or  $S_{\text{qSH}}(p_{xk}, p_{zk})$ .  $g(p_{xk})$  represents  $g_1(p_{xk})$ ,  $g_2(p_{xk})$  or  $g_3(p_{xk})$  based on the respective waves.  $\sqrt{g(p_{xk})}$  is simplified by  $G(p_{xk})$ .

For direct waves, horizontal ray paths  $X^{\text{direct}}$  and travel times  $T^{\text{direct}}$  from the source to the receiver can be written as

$$\begin{aligned} X^{\text{direct}} &= \sum_{k=1}^e \Delta X_k = - \sum_{k=1}^e \Delta Z_k \cdot \frac{\partial G(p_{xk})}{\partial p_{xk}}, \\ T^{\text{direct}} &= \sum_{k=1}^e \Delta T_k = \sum_{k=1}^e \frac{\Delta Z_k}{V_{zk}}. \end{aligned} \quad (7)$$

In reflected waves, ray paths  $X^{\text{reflect}}$  and travel times  $T^{\text{reflect}}$  have similar expressions but with the addition of interface reflection. These results can be easily deduced.

$$\begin{aligned} X^{\text{reflect}} &= \sum_{k=e}^n \Delta X_k^d + \sum_{k=1}^n \Delta X_k^u = - \sum_{k=e}^n \Delta Z_k \cdot \frac{\partial G^d(p_{xk})}{\partial p_x} \\ &\quad - \sum_{k=1}^n \Delta Z_k \cdot \frac{\partial G^u(p_{xk})}{\partial p_x}, \end{aligned}$$

$$T^{\text{reflect}} = \sum_{k=e}^n \Delta T_k^d + \sum_{k=1}^n \Delta T_k^u = \sum_{k=e}^n \frac{\Delta Z_k}{V_{zk}^d} + \sum_{k=1}^n \frac{\Delta Z_k}{V_{zk}^u}, \quad (8)$$

where  $n$  is the layer number of the reflection interface. Terms  $d$  and  $u$  represent *downward* and *upward* directions, as the velocities change in different directions in the VTI media. Following Snell's law in anisotropic media, the horizontal slowness remained constant across different layers. Thus,  $p_{xk}$  can be simplified as  $p_x$ . Detailed analytical formulas for the qP/qSV/qSH waves are provided in Appendix A.

For a given source-receiver horizontal distance,  $X_{\text{true}}$ , the search for ray paths can be formulated as a nonlinear equation (Tian & Chen, 2005):

$$f(p_x) = X_{\text{cal}}(p_x) - X_{\text{true}} = 0, \quad (9)$$

where  $X_{\text{cal}}(p_x)$  is the calculated ray-path horizontal distance from source to receiver using a specific ray parameter (horizontal slowness),  $p_x$ . This value represents  $X^{\text{direct}}$  for direct wave and  $X^{\text{reflect}}$  for reflected wave. We followed the approach of Fang and Chen (2019) to determine the true  $p_x$ . We assumed  $p_x = p_{xi} + \Delta p_x$ , where  $p_{xi}$  is the initial guess and  $\Delta p_x$  is the correction term. The above equation can be expanded using a second-order Taylor series about  $\Delta p_x$ :

$$\begin{aligned} f(p_x) &= f(p_{xi} + \Delta p_x) = f(p_{xi}) + \Delta p_x f'(p_{xi}) \\ &\quad + \frac{\Delta p_x^2}{2} f''(p_{xi}) + O(\Delta p_x^3) = 0. \end{aligned} \quad (10)$$

This leads to a quadratic equation in  $\Delta p_x$ :

$$\begin{aligned} a_1 \Delta p_x^2 + b_1 \Delta p_x + c_1 &= 0, \\ a_1 &= \frac{1}{2} f''(p_{xi}), \\ b_1 &= f'(p_{xi}), \\ c_1 &= f(p_{xi}). \end{aligned} \quad (11)$$

The solution to this quadratic equation is as follows:

$$\Delta p_x = \frac{-b_1 \pm \sqrt{b_1^2 - 4a_1 c_1}}{2a_1}. \quad (12)$$

This equation yields two values for  $\Delta p_x$ . The one that makes the absolute value of  $f(p_x)$  smaller is selected. The expressions of  $[f(p_{xi}), f'(p_{xi}), f''(p_{xi})]$  in coefficients

$[a_1, b_1, c_1]$  are detailed in Appendix B. The accurate calculation of  $p_x$  involves iteratively updating the initial  $p_{xi}$  and correction term  $\Delta p_x$  until the required accuracy is achieved:

$$p_{x(i+1)} = p_{xi} + \Delta p_x. \quad (13)$$

Upon obtaining the final  $p_x$ , the ray paths and travel times in direct and reflected waves can be accurately determined (Equations 7 and 8). Notably,  $p_x$  is within the range  $[-p_{\max}, p_{\max}]$ .  $p_{\max}$  is the maximum value of horizontal slowness  $p_x$  that is calculated when  $p_z = 0$  (Equations 4 and 5). The details of  $p_{\max}$  are shown in Appendix C.

Furthermore, the change rate of the offset with respect to  $p_{xk}$  can be calculated as

$$\begin{aligned} \frac{\partial X^{\text{direct}}}{\partial p_{xk}} &= - \sum_{k=1}^e \Delta Z_k \cdot \frac{\partial^2 G(p_{xk})}{\partial p_{xk}^2}, \\ \frac{\partial X^{\text{reflect}}}{\partial p_{xk}} &= - \sum_{k=e}^n \Delta Z_k \cdot \frac{\partial^2 G^d(p_{xk})}{\partial p_{xk}^2} - \sum_{k=1}^n \Delta Z_k \cdot \frac{\partial^2 G^u(p_{xk})}{\partial p_{xk}^2}. \end{aligned} \quad (14)$$

A higher change rate indicates a greater sensitivity of the horizontal distance to  $p_{xk}$ . The detailed formulas for the qP, qSV and qSH waves are presented in Appendix A. As discussed by Fang and Chen (2019), the change rate approaches infinity as  $p_{xk}$  tends towards its  $p_{\max}$ . Thus, a small variation in  $p_{xk}$  results in a sharp change in the large offsets when  $p_{xk}$  is close to its  $p_{\max}$ . Oversensitivity can result in unstable Newtonian iteration processes and convergence challenges. To address this issue at large offsets, a generalized dimensionless ray parameter,  $q_x$ , is introduced to ensure a fast and stable iterative process. The transformation from  $p_x$  to  $q_x$  is defined as follows:

$$q_x = \sqrt{\frac{p_x^2}{p_{\min}^2 - p_x^2}}, \quad (15)$$

where parameter  $q_x$  has a similar interpretation compared to Fang and Chen (2019).  $p_{\min}$  is the smallest  $M$  maximum horizontal slowness in layers  $[k_1, \dots, k_M]$  and can be expressed as

$$p_{\min} = \min \{p_{\max}^{k_1}, \dots, p_{\max}^{k_M}\} (k_1 < \dots < k_M), \quad (16)$$

where  $M$  represents the total number of ray paths. For direct waves,  $M = e$ . And for reflected waves,  $M = n$  (Figure 1).

In isotropic media,  $p_{\min}$  will be the reciprocal of the highest velocity that the ray passes through (assumed  $k_m$  layer) and  $q_x$  equals to the ratio  $\Delta X_k / \Delta Z_k$  (Figure 1), which is linearly

related to the maximum horizontal offset  $\Delta X_{k_m}$  and is the same as that in Fang and Chen (2019). In anisotropic media,  $p_{\min}$  will be the reciprocal of the highest phase velocity in the  $k_m$  layer and  $q_x$  serves as an equivalent parameter to the ratio  $\Delta X_k / \Delta Z_k$ , as  $\Delta X_{k_m}$  is calculated by the highest group velocity in the  $k_m$  layer. The parameter  $q_x$  is very similar to  $\Delta X_k / \Delta Z_k$ . The offset  $X$  is mainly affected by the maximum offset  $\Delta X_{k_m}$ , which results in the change rate of  $q_x$  decides the change rate of the offset  $X$ . Thus,  $q_x$  has the ability to make slow changes in offset. That will be shown in the numerical examples.

Based on the above equations,  $p_x$  can be replaced with  $q_x$ :

$$p_x = \frac{q_x p_{\min}}{\sqrt{1 + q_x^2}}. \quad (17)$$

This transforms the search for  $p_x$  in Equation (9) into a search for  $q_x$ :

$$f(p_x(q_x)) = X_{\text{cal}}(p_x(q_x)) - X_{\text{true}} = 0, \quad (18)$$

where  $f(p_x(q_x))$  can be abbreviated as  $f(q_x)$ . Assuming  $q_x = q_{xi} + \Delta q_x$ , the above equation can be expanded using a second-order Taylor series about  $\Delta q_x$ :

$$\begin{aligned} f(q_x) &= f(q_{xi} + \Delta q_x) = f(q_{xi}) + \Delta q_x f'(q_{xi}) \\ &+ \frac{\Delta q_x^2}{2} f''(q_{xi}) + O(\Delta q_x^3) = 0, \end{aligned} \quad (19)$$

where  $q_{xi}$  is the initial estimate of  $q_x$ , and  $\Delta q_x$  is the correction increment. The equation is then rewritten as a quadratic equation in  $\Delta q_x$ :

$$\begin{aligned} a_2 \Delta q_x^2 + b_2 \Delta q_x + c_2 &= 0, \\ a_2 &= \frac{1}{2} f''(q_{xi}), \\ b_2 &= f'(q_{xi}), \\ c_2 &= f(q_{xi}). \end{aligned} \quad (20)$$

Details of  $f'(q_{xi})$  and  $f''(q_{xi})$  are presented in Appendix B. The solution to this quadratic equation is as follows:

$$\Delta q_x = \frac{-b_2 \pm \sqrt{b_2^2 - 4a_2c_2}}{2a_2}. \quad (21)$$

The one of  $\Delta q_x$  that reduces the absolute value of  $f(q_x)$  is chosen.  $q_x$  is determined through an iterative process.



$$q_{x(i+1)} = q_{xi} + \Delta q_x. \quad (22)$$

Finally, the true  $q_x$  is inferred using Newton iteration. This transformation from  $p_x$  to  $q_x$  extends the range of the ray parameters from  $[0, p_{\max}]$  to  $[0, \infty)$ , resolving the convergence issues when  $q_x$  tends towards infinity and ensuring stability.

### Initial estimate of $q_x$

In the gradient-based optimization algorithm, the choice of initial value is crucial for achieving fast and accurate results. If the initial value is too large or small, it can get stuck in local minima or fail to converge. The numerical stability of the algorithm depends on how closely the initial value approximates the true one. Tian and Chen (2005) and Fang and Chen (2019) suggested using the values of  $q_x \rightarrow 0$  and  $q_x \rightarrow \infty$  to estimate the initial value of  $q_x$ . To improve the accuracy of the estimate at intermediate offsets, we introduced additional  $q_0 = 1.0$  as the constraint point. Two piecewise functions in the range  $0 \leq q_x < 1.0$  and  $q_x \geq 1.0$  are constructed to approximate the relationship between  $X$  and  $q_x$ . The piecewise functions can be described as

$$\begin{cases} X = \frac{c_1 q_x^2 + c_2 q_x}{c_3 q_x^2 + c_4 q_x + 1} & 0 \leq q_x < 1.0, \\ X = \frac{d_1 q_x^2 + d_2 q_x + d_3}{d_4 q_x + 1} & q_x \geq 1.0. \end{cases} \quad (23)$$

$[c_1, c_2, c_3, c_4]$  and  $[d_1, d_2, d_3, d_4]$  are the coefficients of two piecewise functions. When  $q_x \rightarrow 0.0$ , the numerator in the first piecewise function plays a key role as the denominator reduces to one, just the same as that in Fang and Chen (2019). When  $q_x \rightarrow \infty$ , the second piecewise function can be approximated as a linear function to  $q_x$ , which is similar to Tian and Chen (2005). To maintain stability and values constant at the breakpoint  $q_0 = 1.0$ , we increase the power of  $q_x$  both in numerator and denominator to keep piecewise smoothness. The initial estimation accuracy of  $q_x$  depends on the correct estimation of the coefficients. These coefficients can be determined by equating the numerical values and first/second-order derivatives of our proposed functions with the Taylor expansion formula (Equation 19) at certain points,  $[0, 1.0, \infty]$ . The initial estimate of  $q_x$  is then calculated and defined using the following expression:

$$\begin{cases} q_x = \frac{(c_4 X - c_2) + \sqrt{(c_4 X - c_2)^2 + 4X(c_1 - c_3 X)}}{2(c_1 - c_3 X)} & X < X(1.0), \\ q_x = \frac{(d_4 X - d_2) - \sqrt{(d_4 X - d_2)^2 + 4d_1(X - d_3)}}{2d_1} & X \geq X(1.0). \end{cases} \quad (24)$$

The specific values of coefficients  $[c_1, c_2, c_3, c_4]$  and  $[d_1, d_2, d_3, d_4]$  can be found in Appendix D.

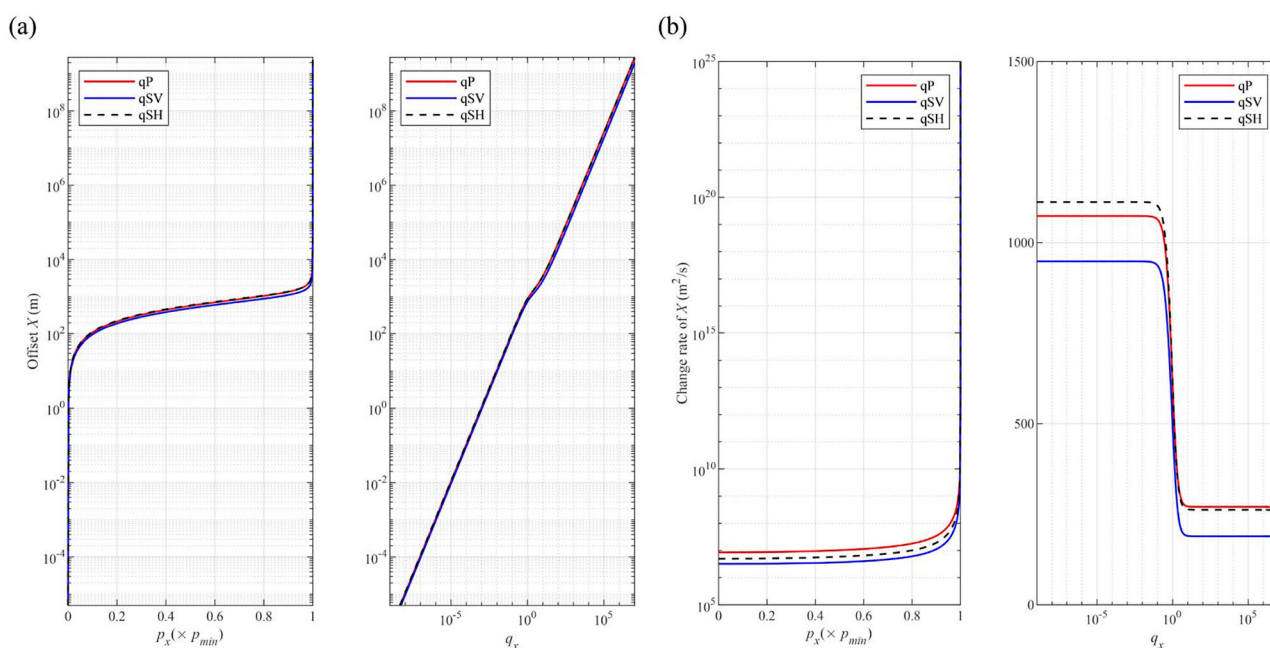
## NUMERICAL EXAMPLE

We created a 10-layer vertical transversely isotropic (VTI) model to illustrate the effectiveness of the  $q_x$  method (Table 1). The layers closer to the surface exhibit weak anisotropy, whereas the deeper layers display strong anisotropy. The specific anisotropic parameters were adjusted based on Koren and Ravve (2018). Parameters  $\alpha_0$  and  $\beta_0$  increased as the number of layers increased, reaching their peak at layer 8 for the stability assessment. Each layer has a uniform thickness of 100 m. In this study, we focused on the reflected wave, considering the direct wave as a part of it. The tenth layer, at a depth of 1000 m, served as the reflector. The source was located at a depth of 10 m, the receivers were positioned on the surface, and different horizontal offsets  $X^{\text{reflect}}$  ( $X$  for short) were tested. Based on the velocity model,  $p_{\min}$  is defined as the  $p_{\max}$  in the eighth layer;  $p_{\max}$  is calculated using the formulae in Appendix C.

First, we compared the performance of the ray parameter  $p_x$  and generalized ray parameter  $q_x$  in relation to different offsets in the reflected wave. Figure 2 illustrates the offset variations and their change rates in logarithmic coordinates under various  $p_x$  and  $q_x$  values. In Figure 2a, the horizontal coordinate of  $p_x$  represents the percentage of the  $p_{\min}$ . Offset  $X$  is relatively short when  $p_x$  is small. The offsets slowly increase as  $p_x$  becomes larger, exhibiting sharp changes when  $p_x$  approaches its maximum value  $p_{\min}$ . Such abrupt changes can easily result in numerical instability, as previously discussed. In contrast, the offsets calculated by  $q_x$  display a more gradual increase regardless of the  $q_x$  value. In Figure 2b, the change rates of the offset provide a direct comparison between  $p_x$  and  $q_x$ . The change rates are significantly higher for smaller  $p_x$ , gradually increasing with larger  $p_x$  values and peaking when  $p_x$  approaches to its maximum,  $p_{\min}$ . A small variation in horizontal slowness  $p_x$  can result in a dramatic change in the offsets. The oversensitivity of the system to small variations in the horizontal slowness can lead to instability in the Newton iteration process. However, the change rates calculated by  $q_x$  remain in a reasonable range when  $q_x$  is small, ensuring that the offsets do not exhibit unexpected fluctuations. Although  $q_x$  is in thousands or more, the change rates decrease to guarantee more stability in the offset change. This stability is crucial for the convergence of the Newton iteration process. The offset can be an approximated linear function of  $q_x$  as it approaches  $\infty$ . This predictability ensures that the Newton iteration process swiftly converges to the true  $q_x$  value across different offsets. More detailed formulae can be found in Appendix D. The offsets exhibit controlled growth regardless of the  $q_x$  value, indicating the ability of the Newton iteration to accurately determine the true  $q_x$  value under varying offset conditions.

**TABLE 1** The anisotropic parameters in the 10-layer vertical transversely isotropic (VTI) model.

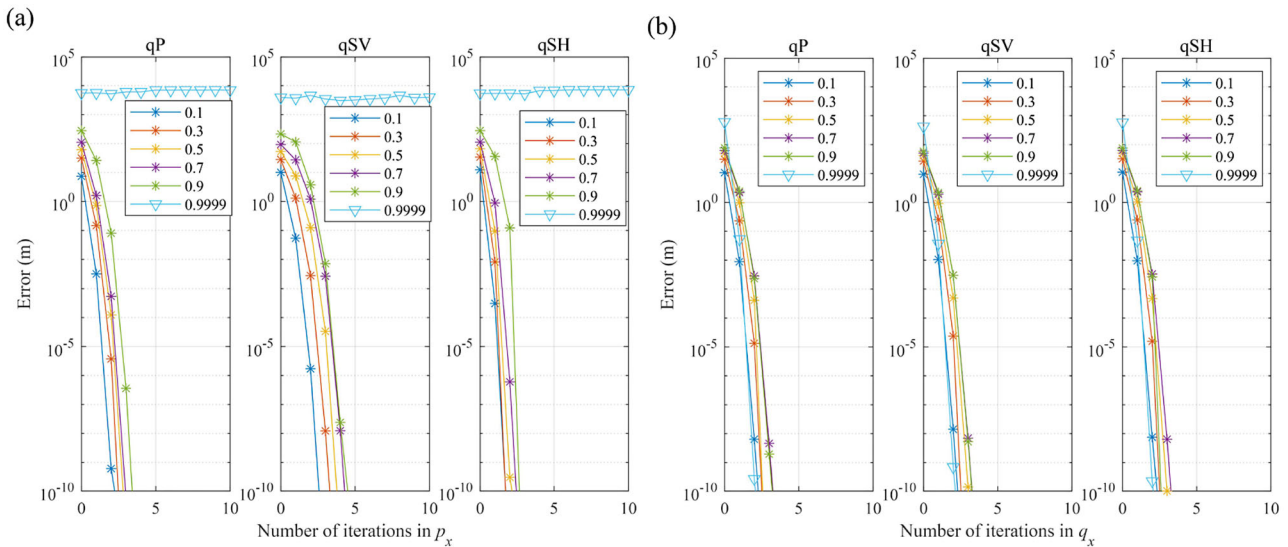
| Layer | $\alpha_0$ (m/s) | $\beta_0$ (m/s) | $\epsilon$ | $\delta$ | $\gamma$ |
|-------|------------------|-----------------|------------|----------|----------|
| 1     | 1500.0           | 1047.0          | 0.03       | 0.04     | 0.02     |
| 2     | 1800.0           | 1123.0          | 0.10       | 0.04     | 0.03     |
| 3     | 2000.0           | 1194.0          | 0.15       | 0.19     | 0.12     |
| 4     | 2100.0           | 1256.0          | 0.25       | 0.20     | 0.25     |
| 5     | 2200.0           | 1379.0          | 0.30       | 0.32     | 0.33     |
| 6     | 2400.0           | 1424.0          | 0.31       | 0.28     | 0.31     |
| 7     | 2500.0           | 1506.0          | 0.26       | 0.28     | 0.36     |
| 8     | 6000.0           | 3423.0          | 0.38       | 0.36     | 0.34     |
| 9     | 3000.0           | 1835.0          | 0.32       | 0.32     | 0.33     |
| 10    | 4000.0           | 2056.0          | 0.33       | 0.36     | 0.35     |

**FIGURE 2** (a) The offsets  $X$  under different  $p_x$  (left) and  $q_x$  (right) reflected from the 10th layer. The horizontal coordinate of  $p_x$  is expressed as a percentage of the maximum value  $p_{\min}$ ; (b) the change rates of offset  $X$  under different  $p_x$  (left) and  $q_x$  (right).

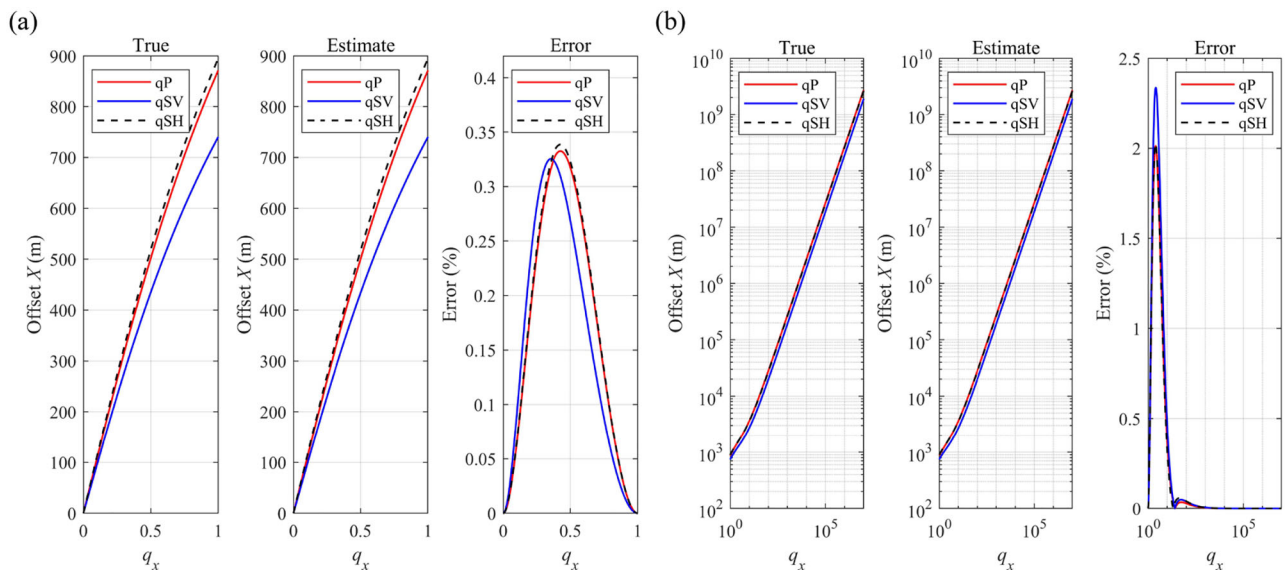
Next, we compare the convergence rates between  $p_x$  and  $q_x$ . Testing offsets are calculated using different values of  $p_x$ , specifically 0.1, 0.3, 0.5, 0.7, 0.9 and 0.9999 times the maximum value,  $p_{\min}$ . The initial values of the Newton iteration are set at 0.9 times the true ones. The misfit function is defined as the absolute difference between the true  $X_{\text{true}}$  and calculated  $X_{\text{cal}}$  in Equation (9). Figure 3 illustrates the iterative misfit errors of the quasi-P (qP), quasi-SV (qSV) and quasi-SH (qSH) waves in the two methods.  $p_x$  leads to convergence errors of less than  $10^{-10}$  m within five iterations for all cases except 0.9999  $p_{\min}$ . This slower convergence is attributed to the high sensitivity of large offsets to  $p_x$ , which can cause Newton iteration failure. In contrast, the  $q_x$  shows

faster convergence rates in a few iterations for various offsets. Specifically,  $q_x$  approaches the true value after only two iterations in the case of 0.9999  $p_{\min}$ . Thus, the generalized ray parameter  $q_x$  demonstrates higher robustness and efficiency than the ray parameter  $p_x$ .

Figure 4 illustrates the accuracy of the estimated offsets using our proposed piecewise functions within the range  $q_x \in [0.0, 1.0]$  and  $q_x > 1.0$ , as determined by Equation (23). Figure 4a,b depicts the true offsets alongside the estimated ones with corresponding percentage errors. Our proposed rational function demonstrates estimated errors below 0.35% within the specific range  $[0.0, 1.0]$ . When  $q_x > 1.0$ , the maximum estimated error remains below 2.5%. Moreover, as



**FIGURE 3** Comparison of the misfit errors between (a)  $p_x$  and (b)  $q_x$ . A test for 0.1, 0.3, 0.5, 0.7, 0.9 and 0.9999 times the maximum value  $p_{\min}$  is conducted. The initial values are taken as 0.9 times the true ones. The figures from left to right are for quasi-P (qP), quasi-SV (qSV) and quasi-SH (qSH), respectively.



**FIGURE 4** The comparison between the true offset and the estimated one in the piecewise range (a)  $q_x \in [0.0, 1.0]$  and (b)  $q_x > 1.0$ . From left to right are the true offsets, the estimated ones and the absolute percentage errors.

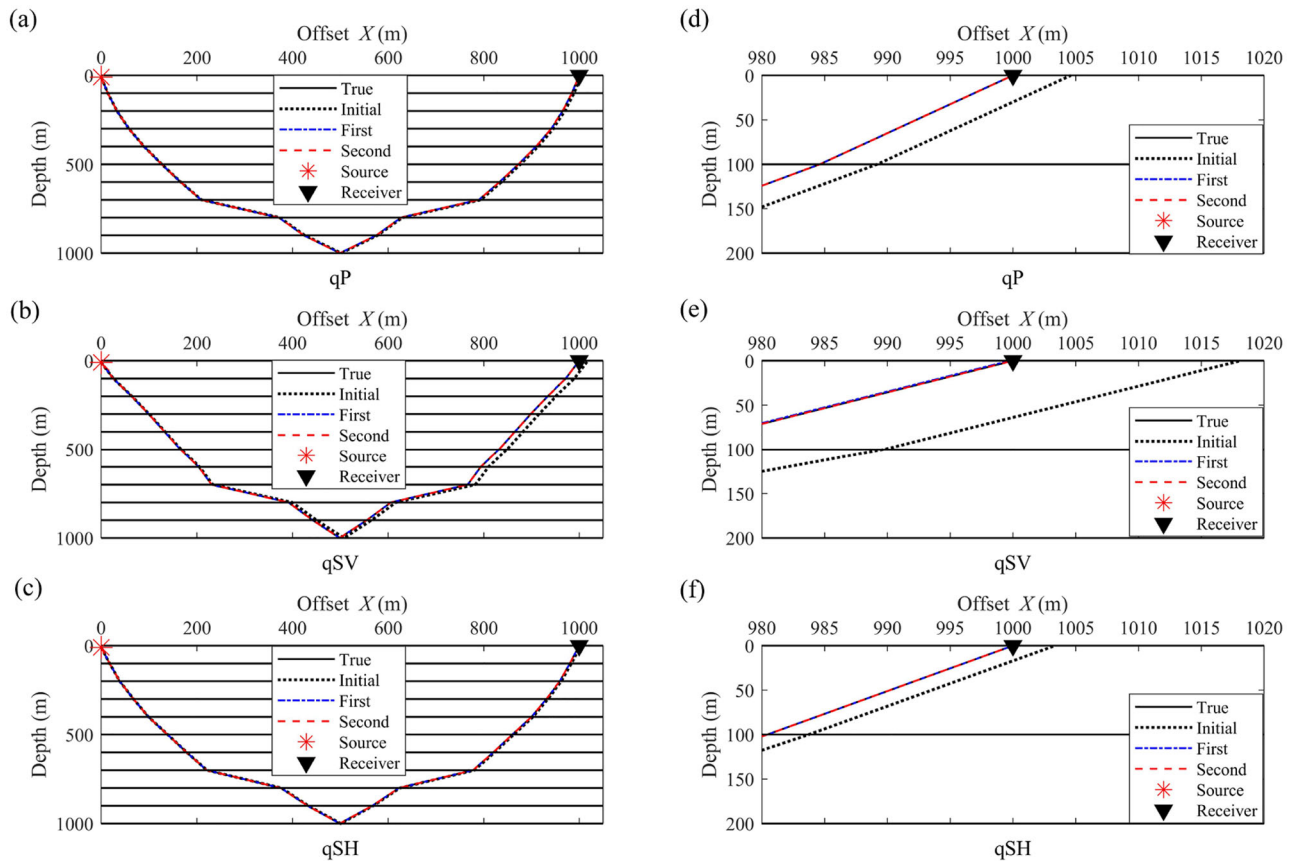
$q_x$  gradually increases, the estimated errors gradually trend towards 0.0. Thus, our approximated functions are highly effective regardless of the value of  $q_x$ . In certain instances, the inferred  $q_x$  from Equation (24) can be directly utilized as the true values.

Figure 5 shows the iterative ray paths originating from our initial estimate of  $q_x$  with an offset of 1000 m. With precise initial estimates, the initial ray paths closely mirror the true ray paths. After one to two iterations, the updated ray paths were

nearly aligned with the true paths. Although accurate initial estimates are beneficial, the method can still rapidly converge to the true values, even with significant initial errors. Nonetheless, a strong initial estimate can provide valuable insight into the model.

Figure 6 shows a comparison between the calculated arrival times and forward waveforms of the reflected qP, qSV and qSH waves. The waveforms were obtained using the generalized reflection/transmission coefficient method (Chen,





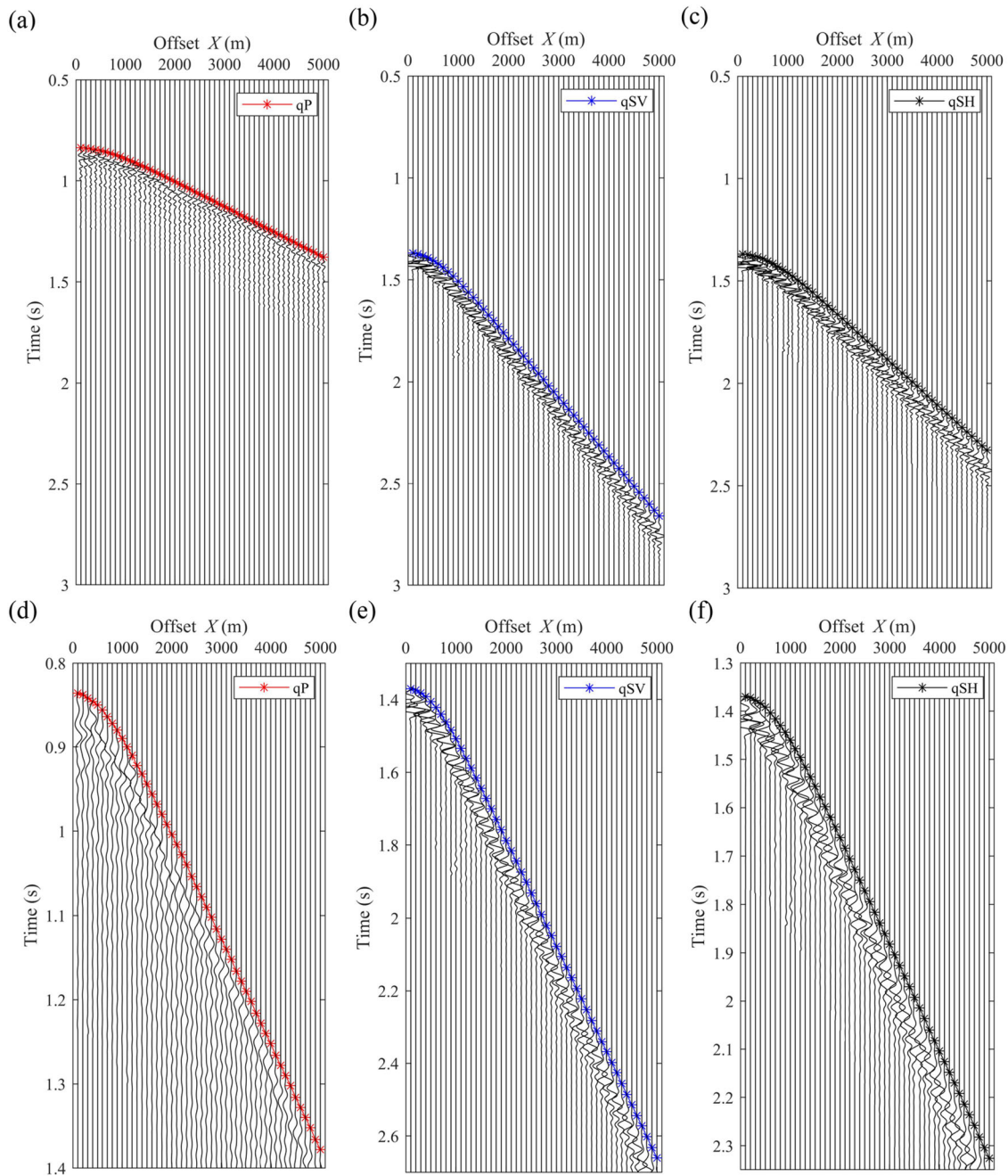
**FIGURE 5** The iterative ray paths starting from our initial estimated  $q_x$  at offset 1000 m in (a) quasi-P (qP) wave, (b) quasi-SV (qSV) wave and (c) quasi-SH (qSH) wave. Parts (d)–(f) are the zoom-in views of the ray paths arriving at the receivers in qP/qSV/qSH waves, respectively. The initial estimated ray paths (black point curves) are very close to the true ones (black solid curves). The first iteration (blue dashed curves) and second iteration (red dashed curves) have almost the same ray paths as the true ones.

1993) with a Ricker wavelet with a 30 Hz centre frequency. Receivers were deployed on the surface at intervals of 100 m, ranging from 100 to 5000 m. Notably, the calculated arrival times aligned flawlessly with the onset times of the forward waveforms, highlighting the exceptional precision of our method.

We conducted two additional tests to assess the robustness and precision of the generalized ray-parameter method. In Figure 7, we show the calculated ray paths for the reflected waves at offsets of 100, 500, 1000 and 5000 m. The three phases, qP, qSV and qSH waves, followed distinct ray paths and encountered different reflection points. Notably, the eighth layer, where  $p_{\min}$  is situated, featured the most extensive horizontal ray paths compared to the other layers, which was particularly evident at larger offsets, such as 5000 m. Unlike  $p_x$ , control parameter  $q_x$  is directly related to  $p_{\min}$  and is crucial for ensuring the robustness of the iterative procedure.

We utilized a Monte Carlo simulation to evaluate the effectiveness of the  $q_x$  method. This test involved 1000 randomly generated models based on the current VTI model.

These models retained the same layer structure but had random depth perturbations within the range of  $\pm 50$  m. The anisotropic parameters,  $[\alpha_0, \beta_0, \epsilon, \delta, \gamma]$ , in each layer were updated using a Gaussian distribution with  $N(0, 0.1)$ . Consequently, the new models featured varying thicknesses of layers and anisotropic parameters. The source remained at a depth of 10 m, with receivers placed on the surface. The reflection of 1000 m at the bottom was constant. In Figure 8, we present the errors with different  $q_x$  values corresponding to offsets of 10, 100 and 1000 km in the 1000 models. The results indicate that errors based on initial estimates may range from meters to hundreds of meters depending on the model. However, these errors rapidly decrease to a magnitude of  $10^{-5}$  m after the first iteration and become less than  $10^{-10}$  m after the second iteration. Although certain models may not reflect real-world conditions, the statistical analysis of the 1000 random models underscores the ability of the  $q_x$  method to achieve rapid convergence at varying offsets. The Monte Carlo simulation affirms the efficiency and accuracy of the proposed method in dealing with complex velocity models.

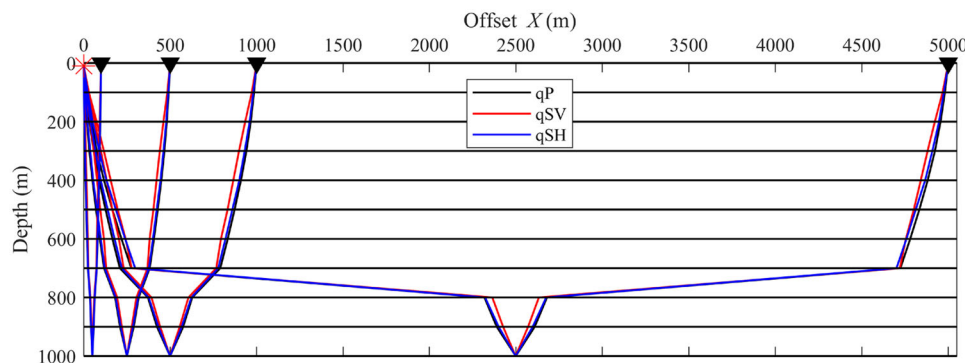


**FIGURE 6** The comparison between our calculated reflected-wave arrival times and the forward waveforms. (a) Quasi-P (qP) wave, (b) quasi-SV (qSV) wave and (c) quasi-SH (qSH) wave. Parts (d)–(f) are the zoom-in views of the arrival times and waveforms in qP/qSV/qSH waves, respectively.

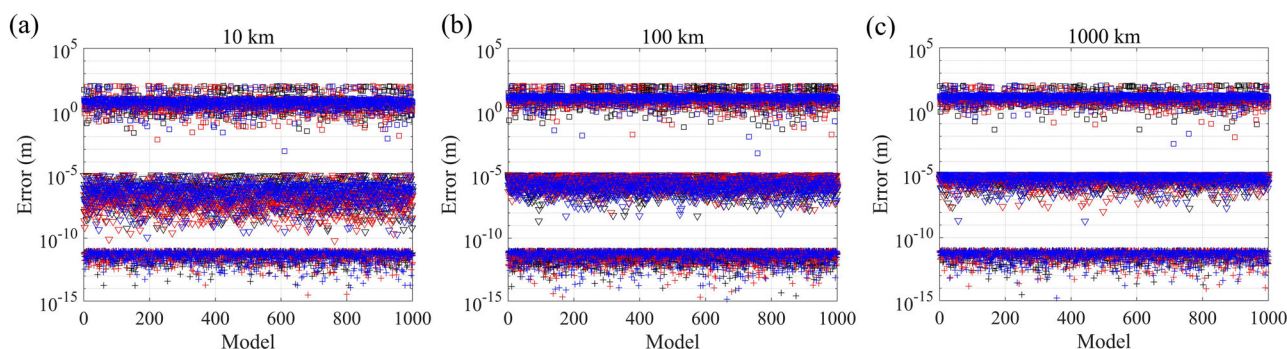
## DISCUSSION

In anisotropic media, the triplication of quasi-SV wave can result in multivalued solutions. Thomsen and Dellinger (2003) explained this issue in terms of the phase and group angles, where the group angle can exhibit multiple values for a single-phase angle. This leads to distinct group-velocity

vertical components with the same horizontal components (Grechka, 2013; Koren & Ravve, 2018). By analysing the relationship between the two components in the slowness domain, we can determine unique solutions under different slowness conditions. Through  $p_x - q_x$  transformation, we can accurately compute the group-velocity components under different  $q_x$  values. As  $q_x$  gradually increases, the two group-velocity



**FIGURE 7** The calculated ray paths for the reflected waves with different offsets responding to 100, 500, 1000 and 5000 m.



**FIGURE 8** The iteration errors for the reflected waves with different  $q_x$  corresponding to (a) 10, (b) 100 and (c) 1000 km in 1000 random models. Squares are the errors for initial estimates, triangles are the errors for the first iteration, and pluses are errors for the second iteration. Black is for the quasi-P (qP) wave, red is for the quasi-SV (qSV) wave, and blue is for the quasi-SH (qSH) wave, respectively.

components maintain unique values. The stability of the  $q_x$  solution is also maintained, thereby avoiding multivalued outcomes.

In our research, we discussed the application of the  $q_x$  method in horizontal layered vertical transversely isotropic media. However, the strata may not always be horizontal owing to tectonic factors, and the anisotropic parameters within the layers may vary based on the depth or angle. All these factors eventually lead to an increased complexity in calculating the ray paths and travel times. We are trying to solve these problems with the  $q_x$  method and will be explored in future research.

## CONCLUSION

Our research presents a fast and robust two-point ray tracing method for strong anisotropic layered vertical transversely isotropic media. Leveraging the Christoffel slowness surface, we developed the  $q_x$  method to efficiently and simultaneously determine the accurate ray paths and travel times for quasi-P, quasi-SV and quasi-SH waves. By modifying horizontal slow-

ness  $p_x$  to generalized ray parameter  $q_x$ , our method converges quickly to true values using the Newton optimization theory. We also proposed a piecewise function for precise initial estimates and demonstrated the effectiveness of the  $q_x$  method in inferring ray paths at large offsets with minimal iterations. A Monte Carlo simulation confirmed its robustness, highlighting its efficiency and accuracy across strong anisotropic media settings.

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## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.



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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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